

Research papers

Extreme precipitation prediction based on neural network model – A case study for southeastern Brazil

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ABSTRACT

Extreme rainfall events can devastate urban and rural infrastructure, affect the economy and even lead to loss of life. In this work, we propose an approach based on Long Short Term Memory networks (LSTM) to forecast precipitation volume extreme rainfall using multivariate time series data. Our methodology combines reanalysis data from 12 isobaric pressure levels, surface data, and data from meteorological stations from Brazil's south-eastern region. Our method allows handles the imbalanced data typical of time-series data used for this type of problem. In order to identify the best model, we performed several experiments with different configurations of LSTM networks. The test results showed that the best prediction model has as input previous data up to 24 h for a forecast of 6 h ahead with a mean absolute error (MAE) of 6.9 mm and root mean squared error (RMSE) of 6.94 mm. Our methodology shows the possibility to use reanalysis data from global mathematical models to obtain less computationally expensive regional models.

1. Introduction

A large amount of the available scientific literature claims that in this century there will be a significant increase in extreme weather events, and consequently extreme rainfall events or, at the very least, an increase in the proportion of heavy rainfall in many areas of the planet (Field et al., 2012; Frame et al., 2020; Lima et al., 2021; Machado et al., 2021). As stated in the Intergovernmental Panel on Climate Change report, titled “Managing the risk of extreme events and natural disasters and promoting adaptations to climate change” (IPCC, 2012), even in regions where there will be a reduction in total annual precipitation, there will be an increase of extreme rainfall events, meaning more heavy rainfall events in a shorter period of time (Field et al., 2012).

Extreme rainfall is a complex event governed by multiple atmospheric (O’Gorman and Schneider, 2009; Duan et al., 2017; Pour et al., 2020; Kodra et al., 2020). Traditionally, weather forecast models are based on mathematical equations to represent the physical laws that govern the movements of the atmosphere and interactions among several different systems that compose the climate system. When using Data-Driven Modelling (DDM), models are not based on primary laws. Instead, they are derived directly from a learning algorithm using data

from previous observations. A data-driven model should reflect the patterns in the training dataset as accurately as possible, but it must also have a high level of generalization, i.e., the ability to provide an adequate prediction for unseen data (Herzog et al., 2018).

A numerical weather forecasting system (Numerical Weather Prediction -NWP) involves a series of sub-processes, such as the collection, modelling, and processing of atmospheric data from surface and altitude weather stations, satellites and weather radars. These systems are used to calculate the atmospheric conditions of a region at a given time interval (Coiffier, 2011). The NWP systems, which combine mathematical models with observations, create datasets for climate monitoring and research, known as reanalysis data. The reanalysis data contains estimates of atmospheric parameters such as air temperature, pressure and wind at different altitudes, and surface parameters, such as precipitation, soil moisture content, height of ocean waves, and temperature of the sea surface. These estimates are produced for all locations on the planet and span several decades.

DDM using artificial intelligence techniques are not entirely new on weather prediction (Geng et al., 2014; Choubin et al., 2016; Ramsundram et al., 2016; Comeaud et al. 2017; Scher, 2018; Samadianfard et al., 2019). In particular, for extreme weather events, several

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approaches have applied artificial intelligence techniques for prediction and classification of extreme rainfall and severe winds (McGovern et al., 2017), predict cyclones, atmospheric rivers, frontal systems (cold / hot) (Liu et al., 2016) and prediction of heat waves (Iglesias et al., 2015). Weather forecasting based on DDM has series of challenges, such as abstraction of spatio-temporal information, high dimensionality, and imbalanced datasets.

In Wang and Ding (2015), the authors proposed a framework for learning patterns of the space-time system and predicting extreme weather events using clustering and classification techniques. Di et al. (2015) proposed an equivalent approach to the previous one, but differentiating extreme and ordinary rainfall events, forming clusters according to the volume of precipitation. After balancing and reducing the dimensionality of the dataset, they applied the K-nearest neighbor algorithm (KNN) to match events to their respective clusters, essentially creating a predictive model. Similarly, Hua and Simovici (2016) implemented a model combining hierarchical clustering to identify locations with similar atmospheric conditions, and used a multivariate time-series data model (Bayesian Structural Vector Autoregression – BSVAR) to make the prediction. Gaitan et al. (2016) used several sources of data such as precipitation data, social and economic data, and data on flood incidents in the Netherlands. They applied clustering to obtain spatially well-defined groups in relation to the volume of precipitation, and were able to predict the impact of floods with a volume of 70 mm (in 6 h).

Yang et al. (2007) segmented sequences of satellite images to extract information from images. Subsequently, they trained a decision tree to classify rainfall events as extreme or ordinary. Lee et al. (2014) used support vector machine (SVM) to classify extreme and ordinary precipitation. In order to address the high dimensionality of the data, the authors employed a feature selection approach based on genetic algorithms. In Kisi et al. (2019), the monthly rainfall amounts were estimated through the SVM and the geostatistical kriging approaches. So, the latitude, longitude, altitude were introduced as model inputs for simulating monthly rainfall values.

Other studies used artificial neural networks (ANN) for rainfall prediction (Phusakulkajorn et al., 2009; Wu and Chau 2013; Taormina and Kwok-Wing Chau, 2015). Phusakulkajorn et al. (2009) transformed the time-series of rainfall volume by wavelet decomposition. This pre-processing decomposes a series of data originally described in the time domain, allowing it to be analyzed in different frequency scales and time periods. After this transformation, they used an ANN to predict the volume of rain for consecutive days. Wu and Chau (2013) combined soft computing approaches for rainfall prediction. The authors proposed preprocessing techniques included moving average (MA) and singular spectrum analysis (SSA) (Ghil et al., 2002). Their results showed that the MA was superior to the SSA when combined with ANN. Taormina and Kwok-Wing Chau (2015) proposed the estimation of prediction intervals (PIs) for operational streamflow forecasting. The method constructs an ANN with two output neurons in order to approximate the lower and upper bounds of the PIs. The authors establish the importance of choosing the proper prediction interval in the design of forecasting models.

Kisi and Shiri (2011) and Azad et al. (2019) proposed models based on neuro-fuzzy techniques for dealing with the non-linear characteristics of the process of precipitation. In Kisi and Shiri (2011), the authors considered wavelet coefficients as inputs in order to represent variations, trends, and periodicities in time series. The results show that the proposed models are effective in forecasting daily precipitation, and simulating monthly rainfall magnitudes, respectively. The main difference between these works and our proposal is that we make predictions of rainfall volume from 6 to 48 h ahead with 6-hour intervals. Furthermore, we consider historical values of atmospheric variables without applying feature extraction processes from the time series. These processes transform original data and can cause the loss of relevant information.

Sulaiman et al. (2018) modelled regional precipitation patterns considering three predictive models based on ANN, SVM, and Random Forest (RF) techniques. For each region, the authors discuss what technique was more efficient to predict the spatial distribution of precipitation in watersheds with limited meteorological data.

Misra et al. (2018) claim that Long Short Term Memory (LSTM) neural networks are capable of capturing a “signature” on the variables at a time before the extreme rainfall event, thus presenting better results than other algorithms. It is worth noting that in that study, there is a complex stage of pre-processing to segment the rainfall into three categories according to its intensity: weak, medium, and strong. The authors also used Principal Component Analysis and decision trees to select the best predictor variables to be used as input to the LSTM neural network. They used the following variables as predictive features: maximum and minimum temperature, mean sea level pressure, specific humidity at 500 hPa, zonal wind velocity, and meridional wind velocity. In contrast to the work proposed in Misra et al. (2018), our predictive variables were eleven surface variables, eight pressure levels variables of 200 to 1000 hPa. The pressure level of 1000 hPa was chosen based on the significant presence of humidity, which can influence the rainfall.

After analyzing the related works, we were able to enumerate the main challenges common to the different approaches:

1. High dimensionality: there is a large number of variables and information available to characterize the rainfall. Due to the fact that data are arranged according to time and space, each data available can be multiplied by many different instances (atmospheric levels, for each latitude, longitude quadrant, etc.). Thus, a weather prediction dataset can easily have thousands or millions of variables¹.

From a data mining perspective, Domingos (2012) points out that high dimensionality is one of the biggest challenges in machine learning and highlights two problems: 1) increase in generalization difficulty as the number of attributes grows; and 2) breaking of thinking based on similarities. Typically, two strategies are proposed to deal with this problem: i) to consider the whole dimension, which demands adequate approaches, techniques, and algorithms for big data scenarios, or ii) to apply a dimensionality reduction process, which demands a feature selection process, e.g. via genetic algorithm. Genetic algorithms and other feature selection approaches, notably, demand great computational efforts. In addition to that, prediction of rainfall volume is a complex, non-linear and multivariable problem. Therefore, any dimensionality reduction strategy can significantly impact the performance of the prediction model. Hence, the participation of domain experts in the feature selection process is important in order to obtain more representative models (Deng et al., 2020).

2. Data imbalance: the imbalance problem is very common in the real world, in cases of extreme rainfall prediction, it is critical. This is because the occurrence of extreme precipitation is much less common than ordinary precipitation. In order to deal with the imbalance, several articles (Wang and Ding, 2015; Di et al., 2015; Wan et al., 2012; Gu and Wan, 2010) used several thresholds (from 90% to 99% of the normal distribution) to distinguish extreme rainfall from ordinary rainfall and then deal with the imbalance, but there is no consensus for this threshold (Stephenson et al., 2008).

3. Spatial resolution of the data versus the scope of the phenomenon: depending on the spatial resolution and temporal evolution of the atmospheric variables, it may be necessary to obtain a greater spatial

¹ The European Centre for Medium-Range Weather Forecasts – ECMWF, (ERA-INTERIM, 2019) collects and provides global weather data every six hours. Considering only the northern hemisphere, there are 320x161 grids and seven isobaric surfaces: 200, 300, 500, 700, 850, 925 and 1000hPa. Each grid has 254 climatic variables. The total number of information fields in climatic data can reach 91,602,560 (320x161x7x254). The ERA-Interim database considers information from several sources, such as: satellites, radio, aircraft, ships, etc.

resolution, in order to accurately describe the phenomenon. Thus, one must consider the capacity of the variables used to represent space–time information and the relationship among them. Segregating and isolating data at a lower resolution may not be realistic, as the dominant atmospheric process may operate at a scale greater than the chosen scale.

4. To deal with the challenges mentioned above, one of the main strategies observed in the literature is differentiating extreme, moderate, and mild rainfall events, and predict precipitation volume according to the predicted severity of the event. Note that this differentiation implies two forecasts, first knowing what level of severity the forecast is at, and then determining the precipitation forecast within that category. In order to increase the reliability of the model, it would be interesting to reduce the number of stages of predictions. Thus, a challenge would be to obtain standalone models for predicting all rainfall events, including extreme and ordinary rainfall.

5. Complexity of models: Most of the proposed models involve the application of various techniques and are strongly dependent on an ordered sequence of procedures which, normally, are difficult to reproduce.

6. Explainability of models: The proposed approaches may result in models with little capacity for interpretation. Understanding the cause and effect conditions of a given phenomenon is often as important as its prediction, and this is the case with extreme rainfall events..

In this work, we propose a data-driven model (DDM) (Minaya et al., 2018) from the exploratory study of reanalysis data, and a methodology based on data mining (DM) techniques, to predict extreme rainfall events. The proposed approach combines reanalysis data from the European Centre for Medium-Range Weather Forecasts, ECMWF (ERA-Interim) (ERA-INTERIM, 2019; Dee et al. 2011), to characterize the dominant processes over conditions of extreme rainfall, seasonality, on a regional scale, with rainfall data from automatic surface stations at the National Institute of Meteorology (INMET) (INMET, 2020). The model aims to identify atmospheric patterns extracted from historical records of extreme rainfall events in order to predict, based on the capture of atmospheric disturbances that occur in previous moments, upcoming rainfall volume.

As a differential of our work, we addressed the aforementioned challenges as follows. With domain knowledge and as suggested by Lima et al. (2010), we selected 12 levels of atmospheric pressure and their predictive variables for rain events provided by the ERA-Interim database. To represent each location in the dataset, we gave the state of these variables between 6 and 96 h earlier prior to the target time-point. This framing of the problem is known as a multi-stage time-series forecasting problem, with the possibility of also predicting several time-points ahead.

In order to handle non-linear characteristics of the atmospheric process, we treated precipitation data as a complete time series, and trained an LSTM neural network capable of capturing cycle characteristics and seasonality of the series. Note that the proposed methodology is able to simplify additional steps, unlike other studies, such as balancing the dataset and reducing the dimensionality and it can be applied to other regions and localities, provided that the target region is fixed. Besides, LSTMs neural networks, an improvement of the recurrent neural networks, are able to capture spatial–temporal dependencies. These nets can preserve long-term information control how much information to store and how much to discard. This characteristic allows dealing with multi-stage time-series forecasting problems, with the possibility of also forecasting several time-points ahead, for extreme and ordinary events.

As a case study, we considered the metropolitan region of Belo Horizonte (MRBH), Minas Gerais State, Brazil (latitude -18.75° to -20.25° and longitude -45° to -43.50°) in the period from 2006 to 2016. In the chosen region, drainage infrastructure is deficient. Even low-magnitude rainfall, if happening over short periods of time (hours), can cause flooding and have catastrophic consequences in the region (Marcelino et al., 2006). This reinforces the relevance of proposing

forecast models for predicting extreme rainfall events (ERE) in this region. The results suggest that climatic variables measured between 96 and 6 h before an ERE- can be used to reliably estimate the volume of precipitation in the near future (6 to 48 h) and, consequently, whether there will be extreme rainfall event. The classification of rainfall in Belo Horizonte shows that rainfall in the range of 30 to 50 mm is considered heavy rain, and rainfall greater than 50 mm is classified as extremely intense (Reis et al., 2004). It is important to emphasize that the proposed methodology is not dependent on a specific study region, although the definition of extreme rainfall events may need to be localized.

2. Methodology: materials and methods

This section describes the considered databases and proposed methodology for building the forecasting model of extreme rainfall events. Fig. 1 shows the flow of the methodology, step 1: definition of the region and period of study; step 2: data collection and extraction procedures; step 3: conceptualization and characterization of the prediction model; step 4: dataset composition (combination ERA-Interim dataset with the precipitation variable from automatic stations) and pre-processing (involving inconsistencies analysis, data normalization, dataset balancing, training and validation processes.); step 5: construction of the LSTM neural model. Finally, step 6: tests and analysis of results. In this last step, we analyze several models considering different forecast horizons and previous historical data. We identified the best models and highlighted their main characteristics. We discuss the performance of our methodology for the prediction of extreme and ordinary rainfall situations and discuss its temporal predictive capacity.

2.1. Materials

2.1.1. Data source from reanalysis database

We used data from the ERA-Interim reanalysis database, collected from 2006 to 2016 (ERA-INTERIM, 2019). The data has a spatial resolution of $0.75^\circ \times 0.75^\circ$ ($\sim 80 \text{ km}^2$), comprising the area between latitude -30° and 0° (41 intervals), and longitude -60° and 0° (81 intervals), totaling 3321 grids that correspond to Brazil's southeastern region. The reanalysis data from ERA-Interim are stored in the Network Common Data Form format (NetCDF3).

For this study, we consider only surface and pressure level variables. Fig. 2 illustrates the grid for data collection at a horizontal resolution of 0.75 in latitude and longitude, with 37 vertical pressure levels, from the surface to 0.1 hPa.

2.1.2. Atmospheric variables from reanalysis

ERA-Interim reanalysis data is available every 6 (six) hours, namely 00:00, 06:00, 12:00, 18:00 UTC - Universal Coordinated Time, of each day. The database provides 254 atmospheric variables for 37 available pressure levels. In our approach, we considered 14 variables (see Table 1). The pressure levels are arranged from the highest isobaric altitude to the lowest altitude in the atmosphere, from 1 to 1000 hPa. Table 2 shows the 77 surface variables available from ERA-Interim database grouped by category.

2.1.3. Data from the surface meteorological stations

The precipitation data recorded on the surface were provided by the Brazil's National Institute of Meteorology (INMET) (INMET, 2020), for the period from 2000 to 2016, recorded in 122 automatic meteorological stations in the southeastern region. Table 3 shows the 17 surface variables, all of these available from INMET database.

2.2. Methods

Fig. 3 shows an infographic detailing the different stage of our methodology (steps: 1–5, see Fig. 1): selection of the target region, data extraction and collection procedures, the conceptualization of

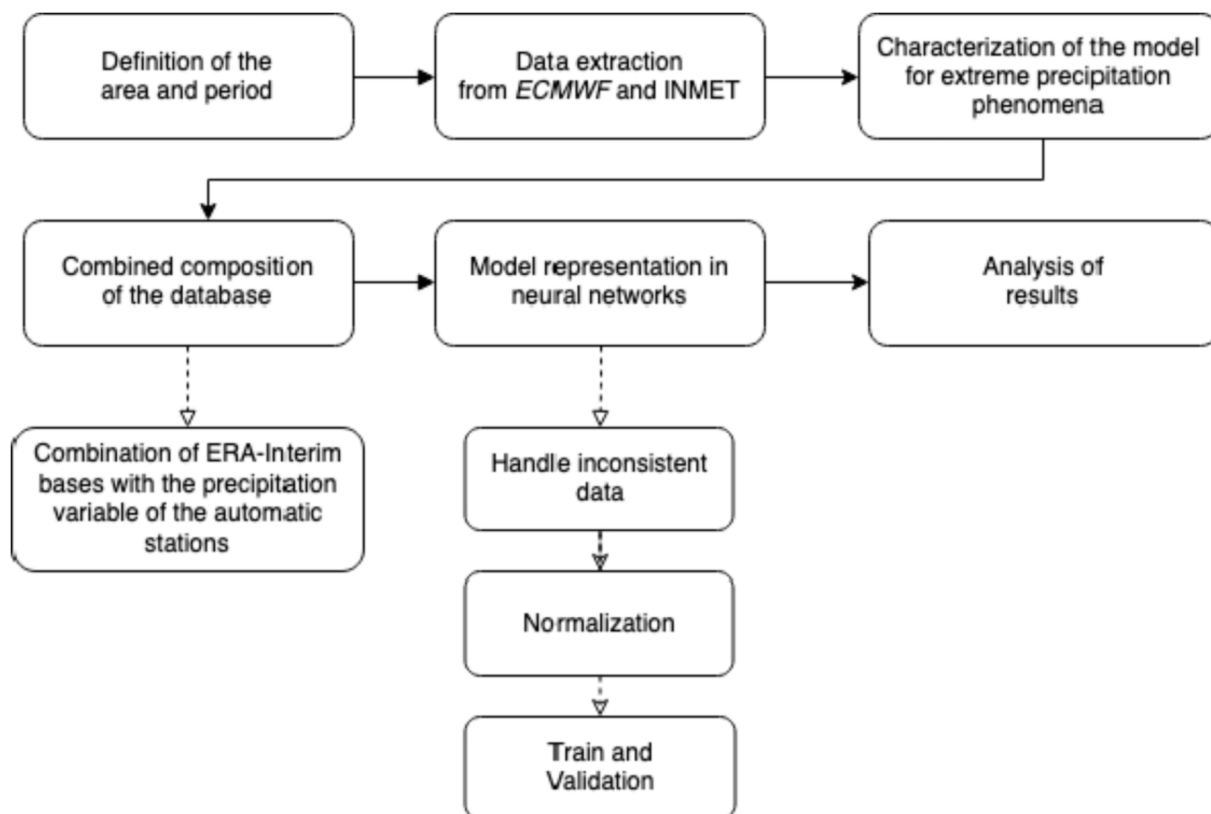


Fig. 1. The Methodology adopted in this study.

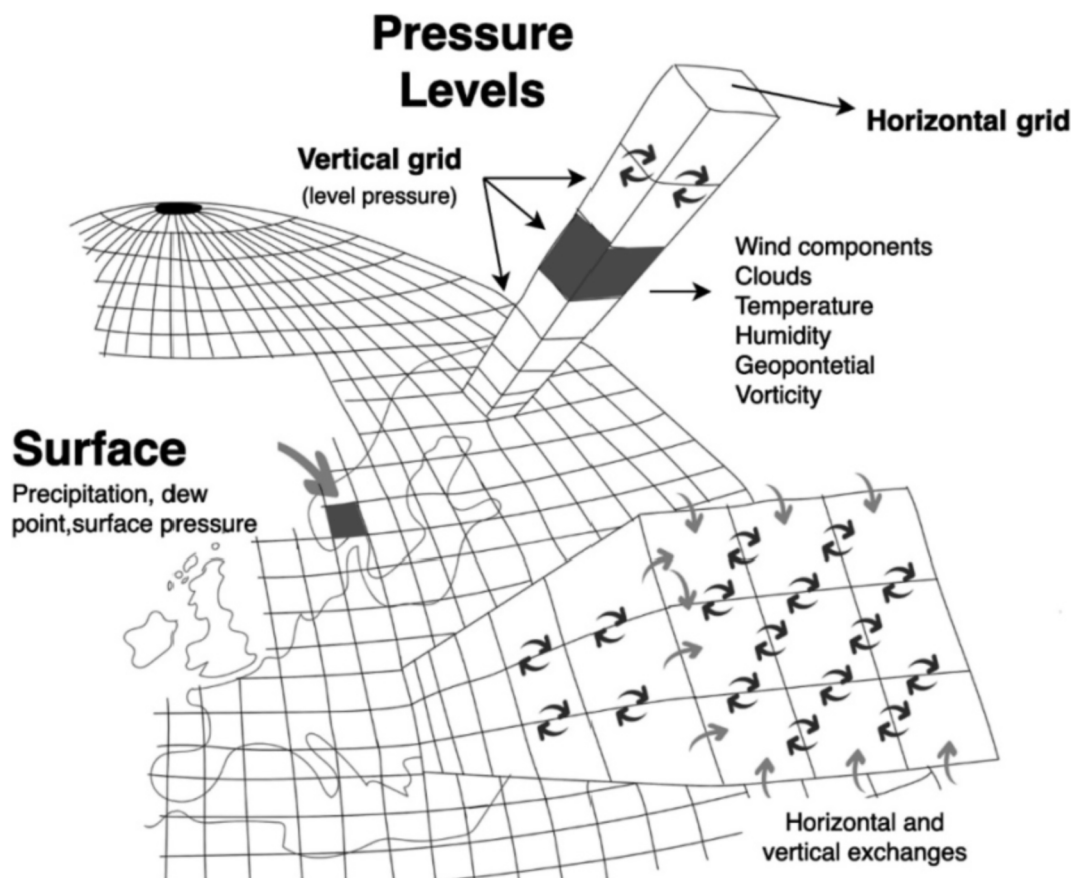


Fig. 2. Horizontal grid of the ERA-Interim model, detailed at each level.

Table 1

Pressure variables provided by the ERA-Interim model.

Pressure levels (hPa)	Variables available for each level	Units
1, 2, 3, 5, 7, 10, 20, 30, 50, 70, 100, 125, 150, 175, 200, 225, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 775, 800, 825, 850, 875, 900, 925, 950, 975, 1000	Potential vorticity	$m^2 s^{-1} K$
	Geopotential	kg^{-1}
	Temperature	$m^2 s^{-2}$
	Specific humidity	$°K$
	Vertical velocity	kg/kg
	Vorticity (relative)	$Pa s^{-1}$
	Divergence	s^{-1}
	U component of wind	s^{-1}
	V component of wind	$m s^{-1}$
	Relative humidity	$m s^{-1}$
	Ozone mass mixing ratio	%
	Specific cloud liquid water content	Kg/kg
	Specific cloud ice water content	Kg/kg
	Fraction of cloud cover	(0–1)

Table 2

Surface variables provided by the ERA-Interim model.

Variables	Units
Sea-ice cover, Snow albedo, Total cloud cover, Albedo, Low cloud cover, Medium cloud cover, High cloud cover	(0–1)
Snow density	$kg m^{-3}$
Sea surface temperature, Ice temperature layer 1, Ice temperature layer 2, Ice temperature layer 3, Ice temperature layer 4, Soil temperature level 1, 2 m temperature, 2 m dewpoint temperature, Soil temperature level 2, Soil temperature level 3, Skin temperature, Soil temperature level 4, Temperature of snow layer	$°K$
Volumetric soil water layer 1, Volumetric soil water layer 2, Volumetric soil water layer 3, Volumetric soil water layer 4	m^3 / m^3
Vertical integral of mass of atmosphere, Vertical integral of water vapour, Vertical integral of cloud liquid water, Vertical integral of cloud frozen water	$kg m^{-2}$
Vertical integral of temperature	$kg m^{-2} K$
Vertical integral of kinetic energy, Vertical integral of thermal energy, Vertical integral of potential + internal energy, Vertical integral of potential + internal + latent energy, Vertical integral of total energy	$J m^{-2}$
Vertical integral of energy conversion, Vertical integral of divergence of kinetic energy flux, Vertical integral of divergence of thermal energy flux, Vertical integral of divergence of geopotential flux, Vertical integral of divergence of total energy flux	$W m^{-2}$
Vertical integral of eastward kinetic energy flux, Vertical integral of northward kinetic energy flux, Vertical integral of eastward heat flux, Vertical integral of northward heat flux, Vertical integral of eastward geopotential flux, Vertical integral of northward geopotential flux, Vertical integral of eastward total energy flux, Vertical integral of northward total energy flux	$W m^{-1}$
Vertical integral of divergence of cloud liquid water flux, Vertical integral of divergence of cloud frozen water flux, Vertical integral of divergence of mass flux, Vertical integral of divergence of moisture flux, Vertical integral of divergence of ozone flux, Vertical integral of mass tendency	$kg m^{-2} s^{-1}$
Surface pressure, Mean sea level pressure	Pa
Snow depth, Skin reservoir content, Surface roughness	m
10 m U wind component, 10 m V wind component	$m s^{-1}$

prediction model, dataset composition, training, and validation of the LSTM model.

2.2.1. Definition of the study region

As case study, we selected the Metropolitan Region of Belo Horizonte, Brazil (MRBH). This region is located between the latitudes of 19°00' and 20°30'S and longitudes of 43°15' and 44°45'W, in the central area of the state of Minas Gerais, in southeast Brazil. Altogether there are 34 municipalities with 9468 km² of extension. The region has about 6 million inhabitants (DSEC, 2015), being Brazil's third most largest urban

Table 3

Variables provided by the automatic meteorological stations.

Variables	Units
Precipitation	mm
Atmospheric pressure	hPa
Maximum atmospheric pressure	hPa
Minimum atmospheric pressure	hPa
Global Radiation	KJ/M^2
Air Temperature	$°C$
Dewpoint Temperature	$°C$
Maximum Temperature	$°C$
Maximum Dewpoint Temperature	$°C$
Minimum Temperature	$°C$
Minimum Dewpoint Temperature	$°C$
Relative humidity	%
Maximum relative humidity	%
Minimum relative humidity	%
Wind speed	m/s
Wind direction	degrees
Wind, Maximum Gust	m/s

area, after São Paulo and Rio de Janeiro's metropolitan regions. In Belo Horizonte city, according to the climatological normal (1981–2010), the pluviometric index is about 1600 mm. per year, distributed almost entirely between the months of October to March. Belo Horizonte's temperature varies on average between 19 °C and 24 °C. The MRBH is very sensitive to flooding due to the pattern of urbanization, inefficient drainage infrastructure, and topography, where altitudes can vary greatly from one region to another. The largest urbanization area is between 751 and 1000 m (altitude), from north to southwest, while the lowest altitudes occur in the northeast, between 650 and 750 m; the largest altitudes occur in the south and southeast limits, between 1001 and 1150 m, reaching 1500 m, at the top of "Serra do Curral". A previous study found a very high correlation of moderate to extreme rainfall with flooding in the MRBH region (Nunes, 2018), hence the relevance of our study.

2.2.2. Data extraction and collection

ERA-InterimThe data of the ERA-Interim were collected in an automated way, in batch, by a routine through an Application Programming Interface (API) (WEBAPI, 2019) available the ECMWF (ERA-INTERIM, 2019) using an access key (previously requested), and parameters such as variables of interest, time period, format, area of interest and resolution. Due to a limit of 20 GB per request, we collected the data over multiple requests (over months) and then merged them into a single dataset.

The files obtained from the ERA-Interim database, in the NetCDF3 format (Conventions: CF-1.6), were stored locally and separated into monthly files spanning 10 years (2006 to 2016), each file has the same spatial resolution of 0.75x0.75, between latitude −30° to 0° and longitude −60° and 0° (Brazil's southeastern region).

3. Meteorological stations – INMET

The weather station data was obtained by digital media requested from INMET (INMET, 2020), stored in files in tabular format, grouped by weather station, and with hourly information for each variable.

3.1. Conceptualisation and characterisation of the forecasting model

Considering the available variables presented in Tables 1, 2 and 3, and being our goal building a model representative for interactions between the pressure and surface variables with the local precipitation target variable, we built a conceptual model, based on explicit and tacit knowledge from domain experts (Lima et al., 2010; Reis et al., 2004).

As mentioned, the ERA-Interim database has 37 pressure levels available. For this study, we selected 12 levels: 200, 300, 400, 500, 600,

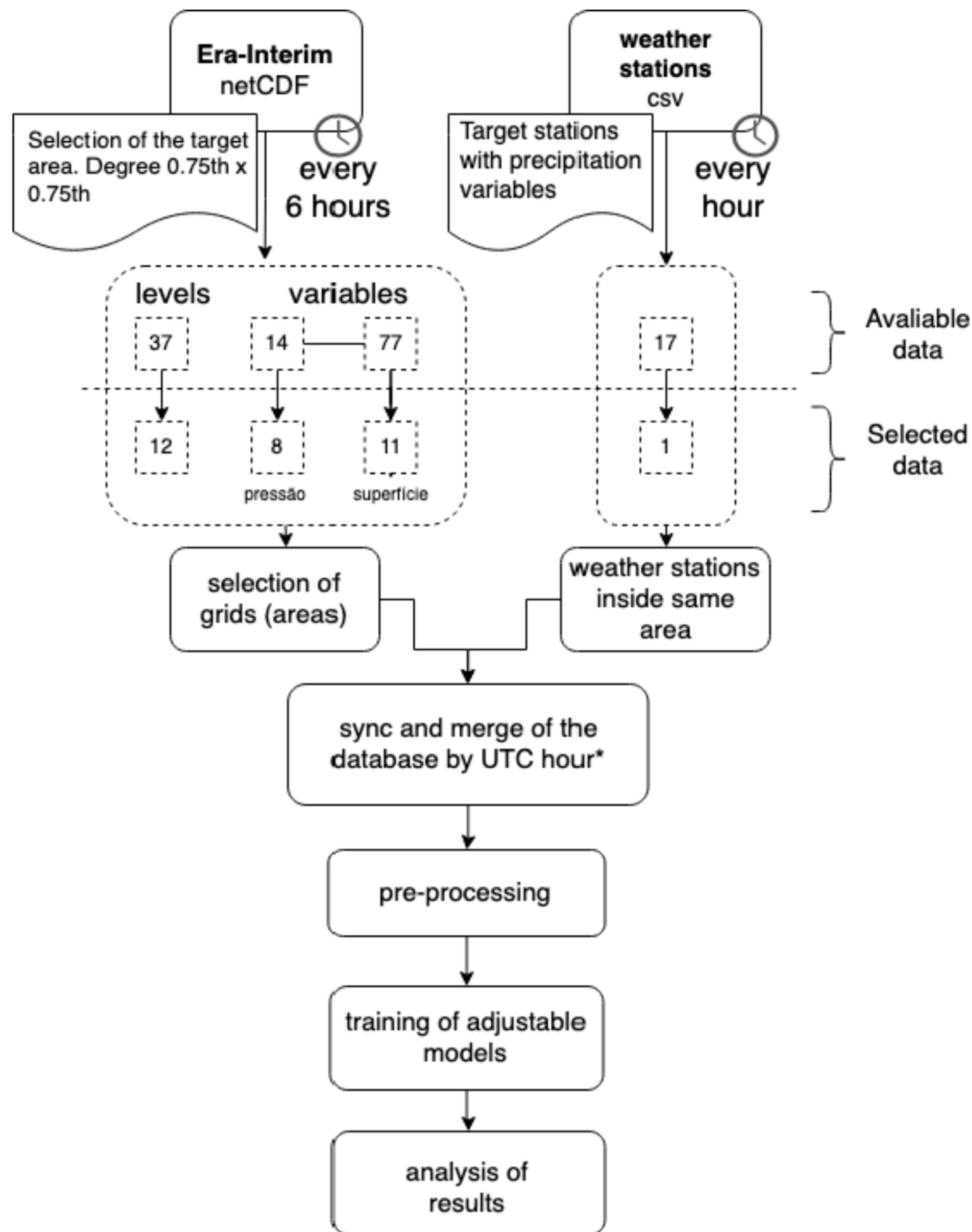


Fig. 3. Infographic: Details of the procedures adopted to obtain the extreme rainfall forecast model.

700, 750, 800, 850, 900, 925 and 1000 (hPa) as suggested in (Lima et al., 2010). The levels selection was based on the significant presence of humidity and wind component that conceptually contribute to the formation of extreme precipitation, and are considered essential predictors. In addition, the values of the variables do not change significantly every 25 hPa, which makes it possible to eliminate intermediate levels. We selected 8 out of 14 pressure variables (Table 1), and 11 out of 77 surface variables (Table 2), all extracted from the ERA-Interim database. The precipitation volume, the target variable, was obtained from INMET's database from automatic surface stations (Table 3). The temporal aspect of the model was incorporated considering the availability of information provided by ERA-Interim, of measurements in 6 h intervals.

ERA-Interim was used to represent the best possible atmospheric data input, which is data from the reanalysis, collected from many

sources and grouped, with observed variables continuously every 6 h. The ERA interim considers the most realistic atmospheric conditions, with very few observation errors. Notably, it can be said that the result of the proposed model has few influences of errors related to the quality of the input data, which are very reliable because are real observations.

3.1.1. Threshold to define an extreme event

The prediction of extreme rainfall events is usually performed according to the considered region for its application (Stephenson et al., 2008). For this, it is necessary to adjust a threshold, in relation to accumulated precipitation in order to define extreme events. In view of this, there are several methodologies to classify extreme precipitation events. e. g. in Field et al. (2012), the authors classify extreme rainfall events as those beyond the 99% percentile of accumulated rainfall, usually greater or

equal to 50 mm rainfall within 24 h. In Di et al. (2015), the authors propose a percentile threshold of 95% within 24 h. Wan et al. (2012), and Gu and Wan (2010) adjusted the threshold in 90%.

Considering the impact of extreme rain to set the threshold, Lee et al. (2014) implemented a model to predict flooding. The authors used a 70.0 mm threshold (in 6 h), based on knowledge from the study's region that this amount of rainfall was capable of causing serious consequences for urbanization. For our case study, we chose a fixed threshold of 50 mm in 24 h, as suggested by Field et al. 2012 and Reis et al., 2004.

The threshold that defines an extreme event may be different for other regions, such as snow-dominated regions, monsoon regions, or droughts regions. The methodology proposed in this work uses the threshold to determine the precipitation volume in 24 h and to identify the atmospheric condition before and during this period. Notice that, this allows us to characterize any rainfall event, extreme or not, independently of the threshold value, making our proposal potentially applicable to other regions and thresholds. As mentioned, our approach uses multivariate regression models as a strategy, and the non-linear aspect, typical of the climatic process, is incorporated by means of a neural network - LSTM.

3.1.2. Characterisation of the prediction model

We formally describe the considered variables set to predict the volume of extreme precipitation as shown in Equation (1).

$$ISPP = \{SFC, PL, Prec\} \quad (1)$$

where *ISPP* corresponds of: surface variables (*SFC*), pressure levels variables (*PL*) and precipitation volume (*Prec*).

The subset of surface variables (*SFC*) corresponds to the variables expressed by Equation (2), where the past time windows are represented by $t-i$.

$$SFC = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_{11} \end{bmatrix}_{(t-n)}, \dots, \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_{11} \end{bmatrix}_{(t-1)}, \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_{11} \end{bmatrix}_{(t)} \quad (2)$$

where:

- s_1 : Total column water ($kg\ m^{-2}$)
- s_2 : Total column water vapour ($kg\ m^{-2}$)
- s_3 : Mean sea-level pressure (Pa)
- s_4 : 10 m U wind component ($m\ s^{-1}$)
- s_5 : 10 m V wind component ($m\ s^{-1}$)
- s_6 : 2 m temperature K
- s_7 : 2 m dewpoint temperature K
- s_8 : Volumetric soil water layer 1 (m^3/m^3)
- s_9 : Volumetric soil water layer 2 (m^3/m^3)
- s_{10} : Volumetric soil water layer 3 (m^3/m^3)
- s_{11} : Volumetric soil water layer 4 (m^3/m^3)

The subset of variables (*PL*) corresponds to the atmospheric variables $PL = \{p_1, p_2 \dots p_8\}$ for different pressure levels (200, 300, 400, 500, 600, 700, 750, 800, 850, 900, 925, 1000), see Equation (3).

$$PL = \begin{bmatrix} p_{200}^1 & p_{200}^2 & \dots & p_{200}^8 \\ p_{300}^1 & p_{300}^2 & \dots & p_{300}^8 \\ \vdots & \vdots & \ddots & \vdots \\ p_{1000}^1 & p_{1000}^2 & \dots & p_{1000}^8 \end{bmatrix}_{(t-n)}, \dots, \begin{bmatrix} p_{200}^1 & p_{200}^2 & \dots & p_{200}^8 \\ p_{300}^1 & p_{300}^2 & \dots & p_{300}^8 \\ \vdots & \vdots & \ddots & \vdots \\ p_{1000}^1 & p_{1000}^2 & \dots & p_{1000}^8 \end{bmatrix}_{(t-1)}, \begin{bmatrix} p_{200}^1 & p_{200}^2 & \dots & p_{200}^8 \\ p_{300}^1 & p_{300}^2 & \dots & p_{300}^8 \\ \vdots & \vdots & \ddots & \vdots \\ p_{1000}^1 & p_{1000}^2 & \dots & p_{1000}^8 \end{bmatrix}_{(t)} \quad (3)$$

where:

- p_{level}^1 - Potential vorticity ($m^2\ s^{-1}\ K\ kg^{-1}$)
- p_{level}^2 - Geopotential ($m^2\ s^{-2}$)
- p_{level}^3 - Temperature (K)
- p_{level}^4 - Specific humidity (kg/kg)
- p_{level}^5 - Vertical velocity ($Pa\ s^{-1}$)
- p_{level}^6 - U velocity ($m\ s^{-1}$)
- p_{level}^7 - V velocity ($m\ s^{-1}$)
- p_{level}^8 - Relative humidity (%)

The precipitation volume (*Prec*) (*mm*) comes from automatic weather stations for a given location at the current time t .

The generic model of precipitation forecast at time t , involving the set of variables expressed by Equations (2) and (3) is given by Equation (4).

$$Prec(t) = \{SFC_{(t-1)}, SFC_{(t-2)}, \dots, SFC_{(t-n)}, PL_{(t-1)}, PL_{(t-2)}, \dots, PL_{(t-n)}\} \quad (4)$$

3.1.3. Dataset composition and preparation

3.1.3.1. Merging the ERA-Interim and meteorological station data. The adopted process for merging the data was to consider the target area before synchronizing the databases. The target location (Belo Horizonte city) is between the longitude -45.0° to -43.5° , and latitude -18.75° to -20.25° corresponding to 9 grids, (3x3) with 0.75x0.75 resolution from ERA-Interim, see Fig. 3.

In order to manipulate smaller files, in this step we built 12 files corresponding to each selected pressure level (200, 300, 400, 500, 600, 700, 750, 800, 850, 900, 925, 1000), containing 8 pressure variables (Equation (3)). Considering the 11 surface variables (Equation (2)) and the target variable (precipitation volume) from the automatic station's data. The datasets with pressure and surface variables were merged according to the dimensions latitude, longitude, time and level, using the merge function of the Pandas library (PANDAS, 2019).

Before to merge the target variable (*Prec*), into the merged dataset, we process the data to accumulate the total precipitation for each 24 h. Fig. 4 illustrates this calculation. The infographic considers a real example: from 16:00 UTC, on 11/28/2006, the INMET Belo Horizonte - Pampulha station, A521, accumulated 73.6 mm rainfall volume in 24 h, which classifies this as an ERE - Extreme Rainfall Event. Assuming the conditions for such an event are detectable at least 24 h before, they were present in the surface and pressure variables, hence the dataset considers the 24 h (UTC) in advance with this upcoming total precipitation. We then added the precipitation (*Prec*) target variable, completing the composition of the dataset.

3.1.3.2. Data preparation. a) Analysing inconsistent records

About 1% of the relative humidity records were outside the expected range: 0.8 to 103, but have not been removed. The support for this decision is as follows: at any temperature and air pressure, a specific maximum amount of water vapor in the air will produce a relative humidity (RH) of 100%. Supersaturated air contains more water vapor than is necessary to cause saturation. The water vapor begins to

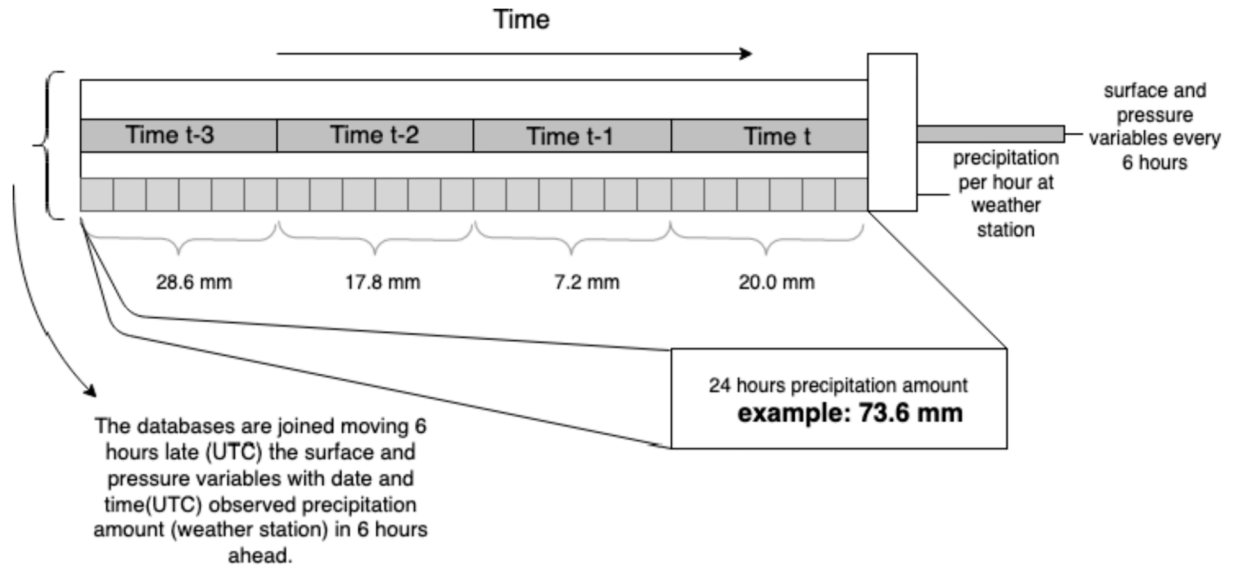


Fig. 4. Diagram describing how the datasets were synchronized.

condense into impurities in the air (such as dust or salt particles), as the relative humidity approaches 100%, and a cloud or mist forms. In the air absolutely devoid of impurities, as is sometimes the case above 9,000 m (but never at lower levels) the UR can rise well above 100%.

b) Data normalisation

A standard LSTM neural network unit consists of a cell, an input gate, an output gate and a forget gate. The cell processes values at arbitrary time intervals and the three gates regulate the flow of information into and out of the cell. To address the vanishing gradient problem in an LSTM, we use a function whose second derivative can sustain for a long time before approaching zero. Typically, in design of LSTM, the sigmoid and hyperbolic tangent functions are commonly used as activation functions in the network units. In this work, we selected both functions that have that property. Since the output gate of an LSTM network has a hyperbolic tangent function, we normalized the data in the $[-1, +1]$ range.

can be accumulated with previous observations such as, $t-1$, $t-2$, $t-3$, $t-n$, (respectively 6, 12, 18 h ago or $6 \times n$ hours ago), to forecast the volume of precipitation at time $t + h$, where $h = 1, \dots, 8$, corresponding to a forecast horizon of 6, 12, ..., 48 h ahead. We trained and evaluated several models, evaluating different temporal windows using the LSTM neural networks: considering the input variables from time $t-16$, $t-12$, $t-8$, $t-4$, $t-2$ and $t-1$; and the predicted variable ($Prec(t+h)$) for the forecast horizons of 6, 12, ..., 48 h ahead. The considered models are presented in Equation (5):

$$Model2 : Prec_{(t+h)} = \{SFC_{(t-2)}, SFC_{(t-1)}, PL_{(t-2)}, \dots, PL_{(t-1)}\} \quad (5a)$$

$$Model2 : Prec_{(t+h)} = \{SFC_{(t-2)}, SFC_{(t-1)}, PL_{(t-2)}, \dots, PL_{(t-1)}\} \quad (5b)$$

$$Model3 : Prec_{(t+h)} = \{SFC_{(t-4)}, SFC_{(t-1)}, PL_{(t-4)}, \dots, PL_{(t-1)}\} \quad (5c)$$

$$Model4 : Prec_{(t+h)} = \{SFC_{(t-8)}, SFC_{(t-1)}, PL_{(t-8)}, \dots, PL_{(t-1)}\} \quad (5d)$$

$$Model5 : Prec_{(t+h)} = \{SFC_{(t-12)}, SFC_{(t-11)}, \dots, SFC_{(t-1)}, PL_{(t-12)}, PL_{(t-11)}, \dots, PL_{(t-1)}\} \quad (5e)$$

c) Data balancing strategy

Because our approach does not separate extreme event records from ordinary events, the dataset is unbalanced by having more instances of ordinary event records than extreme events. In order to balance extreme and ordinary rainfall events, the proposed regression model targets accumulated precipitation over 24 h, so events with accumulated precipitation above 50 mm, for each 6-hour window (time window for changing input variables) are treated as a unique event in the training stage. That is, if a single event lasts for multiple times of 6 h or even days, it will be trained as multiple events of extreme rainfall.

3.1.4. Constructing the LSTM neural models

3.1.4.1. Temporal models for forecasting volume of precipitation. Forecasting the volume of precipitation is intrinsically a time-series analysis problem. Normally, for time-series forecasting, we use lagged observations (e.g. $t-n$) as input variables to forecast in the time interval $(t+h)$. This can be called an h -step ahead forecast or a horizon of h .

For the proposed model (taking into account the range of data availability by the ERA-Interim system) we consider that the time lag

$$Model2 : Prec_{(t+h)} = \{SFC_{(t-2)}, SFC_{(t-1)}, PL_{(t-2)}, \dots, PL_{(t-1)}\} \quad (5f)$$

3.1.4.2. LSTM modeling. The problems involving time-series are more difficult to model. Unlike common regressive predictive modeling, time-series data adds the complexity of a historical sequence dependency between input variables and seasonality.

ANN deal with the nonlinear relationships of complex systems (Bittencourt and Zárate, 2011). Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) are a modified version of recurrent ANN, which have the ability to “forget and remember” the most recent and relevant past. A wide variety of architectures can be designed to an LSTM in order to obtain good predictive models for problems involving time-series (Malakar et al, 2021). Designing LSTM networks is more difficult and there is no well-founded theory on how to do it objectively. Thus, we evaluated our models through several experiments.

Some techniques aim to explore the different configurations using final results as criteria. Grid search is a technique based on the exhaustive search used to find the ideal hyper-parameters, that result in

models with more accurate predictions. The search takes place in a hyper-parameter space defined by a manually specified interval. Typically, a grid search algorithm must be guided by some performance metric, usually measured by the cross-validation technique. In this work, we used the walk-forward validation technique, as it is appropriate for time-series data (Gers et al., 1999).

The mathematical model for LSTMs with forget gates (Gers et al., 1999), considered in this work, is given by Equations (6a) to (6f). The W , U and B matrices are determined through the training process of the neural network.

$$\text{Forgetgate} : F_t = \sigma(W_f \cdot X_t + U_f \cdot H_{t-1} + B_f) \quad (6a)$$

$$\text{Updategate} : I_t = \sigma(W_i \cdot X_t + U_i \cdot H_{t-1} + B_i) \quad (6b)$$

$$\text{Cellcandidatestate} : C_t = \tanh(W_c \cdot X_t + U_c \cdot H_{t-1} + B_c) \quad (6c)$$

$$\text{Outputgate} : O_t = \sigma(W_o \cdot X_t + U_o \cdot H_{t-1} + B_o) \quad (6d)$$

$$\text{Memoryupdate} : C_t = F_t \odot C_{t-1} + I_t \odot C_t \quad (6e)$$

$$\text{Outputgate} : H_t = O_t \odot \tanh(C_t) \quad (6f)$$

Where:

d : number of inputs ($d = 108$);

h : number of hidden units ($h = 300$);

t : timestep index. We use several timesteps in our models;

$\odot$: Hadamard product operator

$X_t \in \mathbb{R}^d$: input vector for the LSTM unit

$F_t \in \mathbb{R}^h$: activation vector for the activation gate

$I_t \in \mathbb{R}^h$: activation vector for the input/update gates

$O_t \in \mathbb{R}^h$: activation vector for the output gate

$H_t \in \mathbb{R}^h$: output vector for the LSTM unit

$C_t \in \mathbb{R}^h$: activation vector for a candidate cell status

$C_t \in \mathbb{R}^h$: cell status vector

$W \in \mathbb{R}^{h \times d}$: synaptic weight matrix

$U \in \mathbb{R}^{h \times h}$: synaptic weight matrix

$B \in \mathbb{R}^h$: polarization weights vector

σ : log-sigmoid activation function

$\tanh$: hyperbolic tangent activation function

3.1.4.3. Strategies for training, validation and testing. When evaluating a model for time series prediction, we are interested in the model's performance on data that was not used during training. In machine learning, we call this a test procedure (with out-of-sample data), which provides a more meaningful assessment. This procedure was carried out by dividing the data into 2 sets, approximately 8 years for training, and 2 years for validation and tests, simply by filtering by the date of occurrence of the events. The evaluation of these forecasts will provide a good demonstration of the model's performance when used operationally.

Approximately 80% of the total dataset (11,656 instances) were used in the training process, corresponding to approximately 8 years, with 4 records per day (00:00, 06:00, 12:00 and 18:00 h (UTC), between 2006 and 2014. For validation and testing, we used the remaining data, a set with 2,920 instances (20% of the total).

In the in time-series forecast, the evaluation of models in historical data is called *backtesting*. As mentioned, for this study we used the *walk-forward* validation technique (Kirkpatrick et al., 2015) where the data is divided into small data segments and chronologically ordered. While in cross-validation, k-folds are defined in which the set will be partitioned, for the *walk-forward* technique, there is no predetermined number. In each training execution, all data available before the target time-point is used in the training set, and the unseen data is used as a validation set. The process is done continuously until the time series ends. Subsequently, the accuracy of the model is computed as the average between

runs. The number of executions varies from 1 up to the total number of predefined executions/segments used for training.

3.1.4.4. Evaluation metrics to the prediction model. Mean absolute error (MAE) and Root mean squared error (RMSE) are two metrics, typically used to evaluate prediction models results based on time-series. MAE corresponds to the mean over the test sample of the absolute differences between prediction and actual observation where all individual differences have equal weight. Meanwhile, RMSE follows a quadratic score rule that also measures the mean magnitude of the error. As RMSE corresponds to the square root of the mean of squared differences between prediction and actual observation will always be greater than MAE for the same results.

A feature of the RMSE is that errors are squared before the mean is calculated. Therefore, the different weights will be added to the sum, and as the error values increase, the RMSE metric increases considerably. As extreme precipitation is an outlier, its weight will be greater for the RMSE calculation and, consequently, it will affect your metric by making it larger. As the main goal of this study aims to predict extreme precipitation among ordinary rains the best approach is to combine RMSE and MAE metrics.

In order to assess the performance of the proposed approach, we carried out a continuous prediction process for historical data involving periods of ordinary and extreme rainfall. The performance of the model for both situations is also analyzed and discussed.

4. Experiments and results analysis

Extreme rainfall events are concentrated within approximately 1% of the normal distribution (Pour et al., 2020; Kodra et al., 2020), which makes it a challenge to obtain a model that can capture these extremes, as well as predicting the volume of precipitation for ordinary rainfall events. Some models choose to separate only the few extreme events, and train a more specific model only for the rainy season or for a specific period. As mentioned, our strategy was to treat the problem as a time-series regression problem, overcoming seasonality (considering periods of low and high rain frequency), and to forecast the volume of precipitation between a window of 6 to 48 h.

In order to evaluate our approach, we analysed the results of training processes considering 11,656 registers from Set/2006 to October/2014. We calculated the RMSE and MAE measures for all neural models for the different time lags (Tables 4 and 5).

Table 4

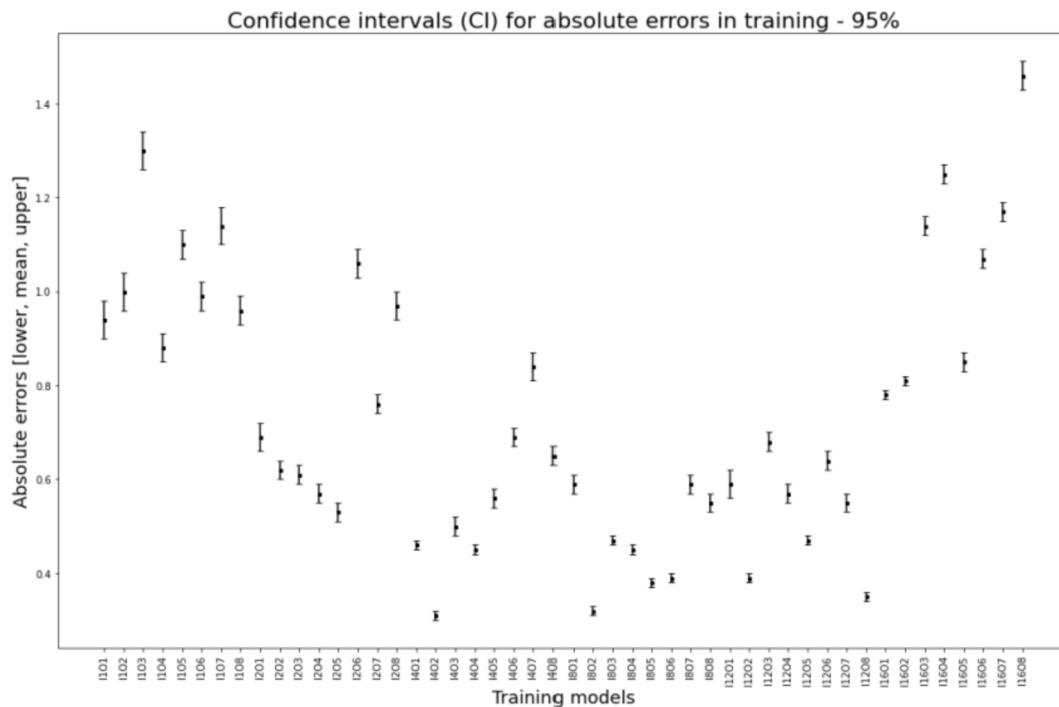
RMSE (mm) for the training process, from Set/2006 to Set/2014.

Models (I)input	RMSE (in mm) (O)output							
	1 (6 hrs)	2 (12 hrs)	3 (18 hrs)	4 (24 hrs)	5 (30 hrs)	6 (36 hrs)	7 (42 hrs)	8 (48 hrs)
1 (6 hrs)	2.19	2.34	2.56	1.64	2.10	2.16	2.39	1.89
2 (12, 6 hrs)	1.59	1.38	1.20	1.23	1.00	1.89	1.55	2.05
4 (24, 18, 12, 6 hrs)	0.94	0.59	1.08	0.77	1.13	1.20	1.81	1.32
8 (48, 42, 36, 30, 24, 18, 12, 6 hrs)	1.23	0.60	0.92	0.88	0.64	0.64	1.04	1.03
12 (72, 66, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	1.70	0.88	1.43	1.14	0.93	1.27	1.06	0.87
16 (96, 90, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	0.88	0.90	1.31	1.41	0.96	1.26	1.33	1.65

Table 5

MAE (mm) for the training process, from Set/2006 to Set/2014.

Models (I)input	Absolute error values MAE (in mm) (O)utput							
	1 (6 hrs)	2 (12 hrs)	3 (18 hrs)	4 (24 hrs)	5 (30 hrs)	6 (36 hrs)	7 (42 hrs)	8 (48 hrs)
1 (6 hrs)	max: 34.79 avg: 0.94 std: 1.97	max: 29.44 avg: 1.00 std: 2.12	max: 25.51 avg: 1.30 std: 2.20	max: 17.22 avg: 0.88 std: 1.38	max: 24.25 avg: 1.10 std: 1.79	max: 33.72 avg: 0.99 std: 1.92	max: 26.67 avg: 1.14 std: 2.10	max: 22.14 avg: 0.96 std: 1.63
2 (12, 6 hrs)	max: 19.74 avg: 0.69 std: 1.43	max: 33.52 avg: 0.62 std: 1.22	max: 11.00 avg: 0.61 std: 1.02	max: 19.02 avg: 0.57 std: 1.09	max: 10.91 avg: 0.53 std: 0.85	max: 23.23 avg: 1.06 std: 1.57	max: 16.60 avg: 0.76 std: 1.35	max: 24.56 avg: 0.97 std: 1.81
4 (24, 18, 12, 6 hrs)	max: 10.61 avg: 0.46 std: 0.82	max: 5.09 avg: 0.31 std: 0.49	max: 13.25 avg: 0.50 std: 0.96	max: 12.80 avg: 0.45 std: 0.62	max: 17.08 avg: 0.56 std: 0.98	max: 27.44 avg: 0.69 std: 0.98	max: 28.81 avg: 0.84 std: 1.60	max: 35.80 avg: 0.65 std: 1.15
8 (48, 42, 36, 30, 24, 18, 12, 6 h)	max: 14.39 avg: 0.59 std: 1.08	max: 6.9 avg: 0.32 std: 0.50	max: 8.27 avg: 0.47 std: 0.79	max: 27.09 avg: 0.45 std: 0.75	max: 7.77 avg: 0.38 std: 0.51	max: 7.26 avg: 0.39 std: 0.51	max: 13.63 avg: 0.59 std: 0.85	max: 19.79 avg: 0.55 std: 0.87
12 (72, 66, ..., 48, 42, 36, 30, 24, 18, 12, 6 h)	max: 63.20 avg: 0.59 std: 1.59	max: 21.47 avg: 0.39 std: 0.78	max: 40.15 avg: 0.68 std: 1.25	max: 12.36 avg: 0.57 std: 0.98	max: 34.45 avg: 0.47 std: 0.81	max: 18.99 avg: 0.64 std: 1.09	max: 29.37 avg: 0.55 std: 0.90	max: 34.41 avg: 0.35 std: 0.79
16 (96, 90, ..., 48, 42, 36, 30, 24, 18, 12, 6 h)	max: 23.21 avg: 0.39 std: 0.78	max: 19.59 avg: 0.38 std: 0.81	max: 28.25 avg: 0.63 std: 1.14	max: 17.14 avg: 0.66 std: 1.25	max: 32.10 avg: 0.46 std: 0.85	max: 31.97 avg: 0.66 std: 1.07	max: 33.79 avg: 0.63 std: 1.17	max: 26.57 avg: 0.74 std: 1.46

**Fig. 5.** The confidence interval for the mean of MAE with confidence level of 95%. Results obtained during training process.**Table 6**

RMSE (mm) for the testing process, from Oct/2014 to Oct/2016.

Models (I)input	RMSE (in mm) (O)utput							
	1 (6 hrs)	2 (12 hrs)	3 (18 hrs)	4 (24 hrs)	5 (30 hrs)	6 (36 hrs)	7 (42 hrs)	8 (48 hrs)
1 (6 hrs)	8.02	9.57	9.33	10.96	10.65	11.17	10.93	11.66
2 (12, 6 hrs)	7.58	9.33	9.39	11.38	10.07	10.36	11.35	11.33
4 (24, 18, 12, 6 hrs)	6.94	8.72	9.53	10.91	10.90	11.28	11.33	11.49
8 (48, 42, 36, 30, 24, 18, 12, 6 hrs)	7.79	8.74	10.04	10.57	11.16	11.06	11.58	12.03
12 (72, 66, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	7.92	9.03	10.26	10.65	10.88	11.19	11.36	12.22
16 (96, 90, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	11.27	9.25	9.75	10.59	11.28	11.69	11.87	12.00

From the results shown in Table 4, it is possible to observe that all models presented satisfactory results. The minimum RMSE value was 0.59 mm for the I4O2 model (Input 4 and Output 2), and the maximum value was 2.39 mm for the I1O7 model (Input 1 and Output 7). Table 5 shows the mean absolute error (MAE) for all models considered in Table 4. Note that the mean errors for each model correspond to values that can be considered low (minimum mean of 0.31 mm for I4O2 and maximum mean of 1.30 mm for I1O3).

Although the maximum MAE value has reached, in some cases, values close to 30 mm, it is important to emphasize that the standard deviation remained at low values. For example, for the I4O2 model, the MAE value was 0.31 mm, and standard deviation of 0.49 mm. Considering a range around the mean and 2 times the standard deviation, we can establish that 95.46% of the error values are within the range between 0 and 1.29 mm. This allows us to conclude that the models are highly representative of the training set.

In order to infer the performance of the models for new data samples, we calculated the confidence intervals for the 95% confidence level (Fig. 5). It can be seen that the performance of the models is satisfactory for all configurations. The mean MAE (mean absolute error) varies between 0.31 and 1.65 mm, and the confidence intervals range from 0.02 to 0.08 mm.

As is well known, computational models based on data always present better results during the training process. However, when compared to performance during its operational use, performance typically decreases. In order to evaluate the models under operational conditions and identify the best prediction model, a test process was carried out in the proposed models.

We evaluated the models using testing data, contains 2920 registers (730 days $\times$ 4 temporal windows of 6 h each one), from October/2014 to October/2016, not considered during training process. We calculated the RMSE of all neural network configurations (Table 6), as well as MAE, and the standard deviation for the different time lags (Table 7).

The best model (Input 4 and Output 1 – I4O1, in bold) has RMSE value of 6.94 mm, see Table 6 (in bold), MAE value of 2.97 mm, and standard-deviation of 6.2 mm for the test data, as shown in Table 6. This model was trained with input from the previous 24 h (24, 18, 12, 6 h) to predict the volume of precipitation 6 h ahead. The second and third best models are I2O1 and I8O1, which have a RMSE value of 7.58 mm. (MAE of 2.98, standard-deviation of 6.9) and 7.79 mm. (MAE of 3.30, standard-deviation of 6.9 mm) respectively. From these three models, it

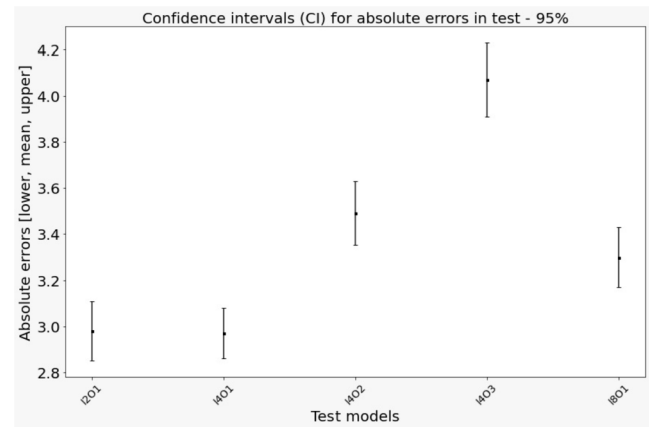


Fig. 6. The confidence interval for the mean of MAE with confidence level of 95%. Results obtained during testing process.

is possible to conclude that the better prediction corresponds to 6 h ahead (O1), considering the previous atmospheric conditions until 48 h (I8).

It is important to highlight that the proposed methodology also presents satisfactory results to the forecasting of 12 and 18 h ahead, see Tables 6 and Table 7, in bold, (I4O2: RMSE = 8.72, MAE = 3.49; I4O3: RMSE = 9.53, MAE = 4.07). These results are important because they can allow guiding anticipated preventive actions to avoid human and economic losses from eminent extreme events.

In order to infer the performance of models (I2O1, I4O1, I8O1, I4O2, I4O3) considering testing data, not used during training processes, we calculated the confidence intervals for the 95% confidence level (Fig. 6). Notice that, the performance of the models is satisfactory. The amplitude of the confidence intervals varies from 0.23 mm (I4O1) to 0.31 mm. (I4O3).

It is important to highlight that our methodology allows predicting extreme rainfall events and ordinary rainfall. In order to evaluate the performance of our approach in both rainfall situations, Fig. 7 shows the Predicted versus Actual rainfall volume values, from the models with lower mean absolute errors - models with inputs 1, 2, 4 and 8, (6, 12, 24 and 48 h ago, respectively) and forecasting of 6 h ahead. Note that the amount of data relating to low rainfall conditions is noticeably greater

Table 7
MAE (mm) for the testing process, from Oct/2014 to Oct/2016.

Models	Absolute error values MAE (in mm)							
	(O)utput							
(I)ntput	1 (6 hrs)	2 (12 hrs)	3 (18 hrs)	4 (24 hrs)	5 (30 hrs)	6 (36 hrs)	7 (42 hrs)	8 (48 hrs)
1 (6 hrs)	max: 113.25 avg: 2.79 std: 7.5	max: 79.61 avg: 3.95 std: 8.7	max: 78.43 avg: 3.71 std: 8.5	max: 93.97 avg: 4.74 std: 9.8	max: 106.43 avg: 4.37 std: 9.7	max: 123.86 avg: 4.70 std: 10.1	max: 97.62 avg: 4.50 std: 9.9	max: 95.12 avg: 5.17 std: 10.4
2 (12, 6 hrs)	max: 82.18 avg: 2.98 std: 6.9	max: 79.96 avg: 3.77 std: 8.5	max: 83.45 avg: 3.78 std: 8.5	max: 93.70 avg: 4.89 std: 10.2	max: 86.12 avg: 4.40 std: 9.0	max: 86.04 avg: 4.44 std: 9.3	max: 97.35 avg: 4.65 std: 10.3	max: 94.62 avg: 4.92 std: 10.2
4 (24, 18, 12, 6 hrs)	max: 82.96 avg: 2.97 std: 6.2	max: 82.51 avg: 3.49 std: 7.9	max: 85.89 avg: 4.07 std: 8.6	max: 96.58 avg: 4.64 std: 9.8	max: 93.83 avg: 4.48 std: 9.9	max: 128.43 avg: 5.00 std: 10.1	max: 95.97 avg: 4.90 std: 10.2	max: 97.44 avg: 4.76 std: 10.4
8 (48, 42, 36, 30, 24, 18, 12, 6 hrs)	max: 86.12 avg: 3.30 std: 7.0	max: 80.98 avg: 3.80 std: 7.8	max: 85.98 avg: 4.52 std: 8.9	max: 97.55 avg: 4.38 std: 9.6	max: 82.47 avg: 4.98 std: 9.9	max: 86.04 avg: 4.84 std: 9.94	max: 97.17 avg: 5.02 std: 10.4	max: 96.38 avg: 5.63 std: 10.6
12 (72, 66, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	max: 80.76 avg: 3.41 std: 7.1	max: 85.87 avg: 3.92 std: 8.1	max: 86.07 avg: 4.37 std: 9.2	max: 93.32 avg: 4.51 std: 9.6	max: 96.35 avg: 4.49 std: 9.9	max: 96.13 avg: 4.52 std: 10.2	max: 96.50 avg: 4.80 std: 10.2	max: 97.21 avg: 5.50 std: 10.9
16 (96, 90, ..., 48, 42, 36, 30, 24, 18, 12, 6 hrs)	max: 97.80 avg: 3.65 std: 10.6	max: 85.66 avg: 3.98 std: 8.35	max: 86.03 avg: 4.19 std: 8.79	max: 97.26 avg: 4.22 std: 9.7	max: 97.40 avg: 4.94 std: 10.1	max: 96.26 avg: 4.98 std: 10.5	max: 96.42 avg: 5.09 std: 10.7	max: 96.91 avg: 5.03 std: 10.8

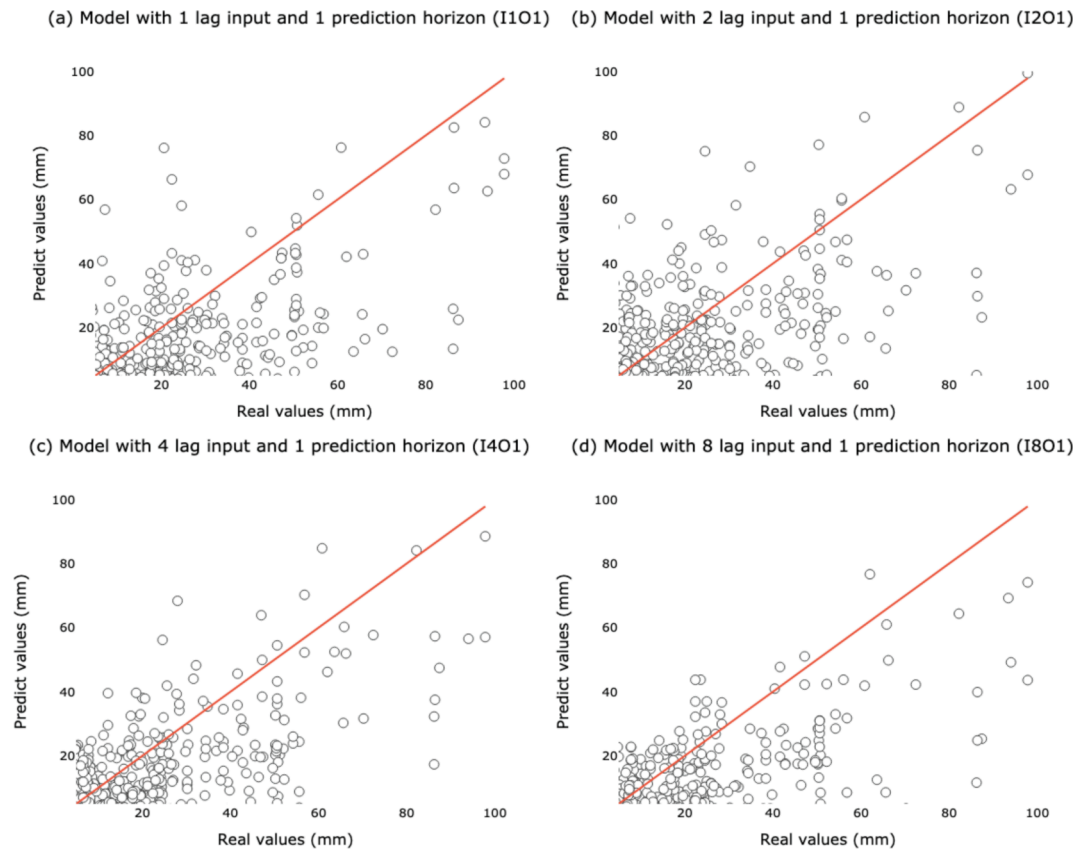


Fig. 7. Prediction vs real values in mm of models with the lowest mean errors. Models with input 1, 2, 4 and 8 (6, 12, 24 and 48 h ago, respectively) and forecast for 6 h.

Table 8

Number of events with rainfall grouped by 10 mm bands from 2014 to 2016.

10 mm* group	Number of observed events	Number of predicted events
(1, 10]	264	556
(10, 20]	147	180
(20, 30]	87	96
(30, 40]	34	29
(40, 50]	22	18
(50, 60]	28	14
(60, 70]	6	6
(70, 80]	2	0
(80, 90]	6	1
(90, 100]	4	1
(100, 110]	0	1
(110, 120]	0	0

*Except the first interval.

than the amount of data relating to extreme event conditions. Despite the applied balancing procedure (subsection 2.2.4.2c), it is expected that the models have a better prediction performance for low rainfall than for high rainfall events. We performed additional experiments considering only data of extreme rainfall. However, due to the small amount of data, the models did not perform any better. The reason why we kept our forecast considering the complete time series.

It is important to emphasize that models based on neural networks are fitted from data and the more information is provided during the training process, these networks are able to represent the qualitative relationships existing in the physical system as exposed in [Bittencout and Zárate \(2011\)](#). So, we consider that modeling the climatic system as a time-series regression problem preserves the greatest amount of information. That is why we chose to work with global models to forecast precipitation regardless of low or high precipitation.

The value of the Pearson correlation coefficient (R^2), considering ordinary and extreme rainfalls, for the four models considered I1O1, I2O1, I4O1, I8O1 are: 0.71, 0.73, 0.72 and 0.70 respectively, which can be considered with high correlation ([Hinkle, et al. 2003](#)). Considering the best model (I4O1), which obtained a RMSE value of 6.94 mm, the Pearson correlations value, considering rainfall events low and up of 50 mm. were of 0.60 and 0.59 considered moderate correlation.

[Table 8](#) shows the number of events with rainfall for the study region, specifically, at the location of the A521 station, latitude: $-19^{\circ}53'2.22''$; longitude: $-43^{\circ}58'9.83''$, altitude 854 m. Notice that, of the 2920 records, 600 rainfall events, above 1 mm in 24 h, occurred. The model predicted 902 rainfall events above 1 mm, being that the biggest difference occurs for the interval (1, 10], where the number of predicted events were of 556 in relation to 264 actual events. The difference is due to the prediction of actual events below and close to 1 mm for which the model predicted values between (1, 10].

It is still possible to note that in 2 years (2014–2016) there were 46 extreme rainfall events (≥ 50 mm in 24 h in bold) corresponding to 7.6% of the total of events that occurred in the considered period. The best model (I4O1) was able to predict the precipitation value for 23 extreme events (50% of the total events). We emphasize that the important thing for prediction is to know if an event will happen, and if it will cause an impact, in relation to the volume of precipitation. We also observed that, for the (50, 70] interval, which characterizes extreme events with socio-economic impact, our model correctly predicted 30 of the 34 events.

Besides, in [Table 8](#), we can highlight that the model (I4O1) is also capable of predicting the value of precipitation for events of lesser intensity. This increases the relevance of the model, e.g. to predict the amount of rain for agricultural systems, which can benefit from this information for planning irrigation periods. As already mentioned, due to a large amount of data referring to conditions of low precipitation, the

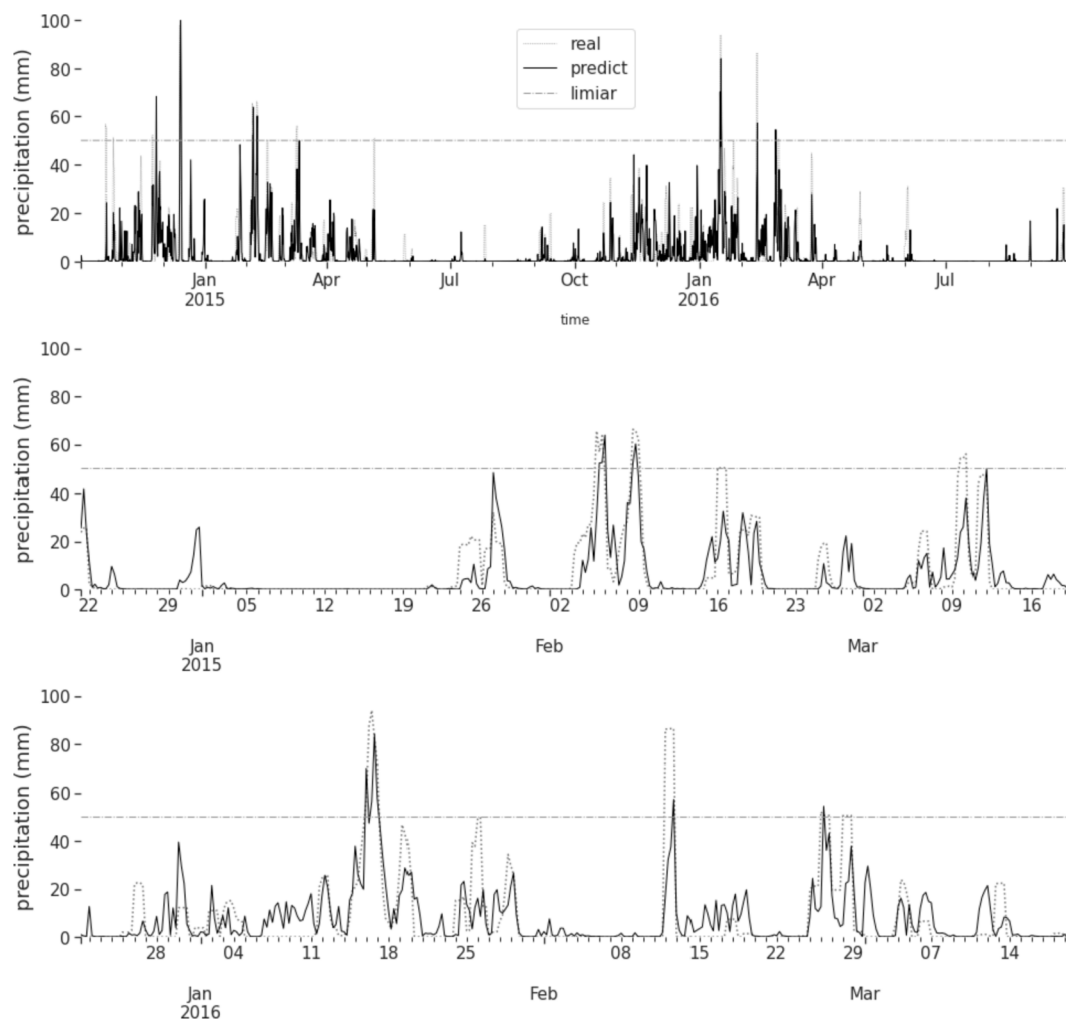


Fig. 8. Prediction for historical data involving periods of normal and extreme rainfall. At the top can be seen a zoom-in of extreme events between Jan 2015 and Mar 2015, at the middle events between Jan 2016 and Mar 2016, and at the bottom is the entire test period from Oct/2014 to Oct/2016 in Brazilian southeastern.

Table 9

Extreme rainfall events from 2014 to 2016 – I4O1 model.

Date-time	Real (mm)	Forecast (mm)	Absolute error	Date-time	Real (mm)	Forecast (mm)	Absolute Error
2014-10-20 6:00:00	56.6	16.2	40.4	2015-03-10 0:00:00	54.2	39.2	15.0
2014-10-20 12:00:00	55.6	36.5	19.1	2015-03-10 6:00:00	56	48.6	7.4
2014-10-20 18:00:00	55.6	42.7	12.9	2015-05-06 0:00:00	50.8	4.3	46.5
2014-10-21 0:00:00	55.6	31.1	24.5	2015-05-06 6:00:00	50.8	33.4	17.4
2014-10-26 0:00:00	51.2	14.6	36.6	2015-05-06 12:00:00	50.8	37.7	13.1
2014-11-23 0:00:00	52	26.3	25.7	2015-05-06 18:00:00	50.4	45.7	4.7
2014-11-23 6:00:00	52.2	33.1	19.1	2016-01-16 0:00:00	56.8	50.9	5.9
2014-12-13 6:00:00	70.2	18.9	51.3	2016-01-16 6:00:00	87.4	47.7	39.7
2014-12-13 12:00:00	97.8	80.9	16.9	2016-01-16 12:00:00	94	61.1	32.9
2014-12-13 18:00:00	97.8	109.2	11.4	2016-01-16 18:00:00	82.2	61.8	20.4
2014-12-14 0:00:00	93.4	96.1	2.7	2016-01-17 0:00:00	72.4	26.7	45.7
2014-12-14 6:00:00	60.8	61.6	0.8	2016-02-12 0:00:00	86.2	3.2	83.0
2015-02-05 6:00:00	65.6	24.6	41.0	2016-02-12 6:00:00	86.2	57.1	29.1
2015-02-05 12:00:00	56.8	38.3	18.5	2016-02-12 12:00:00	86.4	56.2	30.2
2015-02-05 18:00:00	63.6	50.1	13.5	2016-02-12 18:00:00	86.4	61.6	24.8
2015-02-08 12:00:00	66.2	38.6	27.6	2016-02-26 0:00:00	52.2	2.5	49.7
2015-02-08 18:00:00	65.8	63.3	2.5	2016-02-26 6:00:00	50.6	38.9	11.7
2015-02-09 0:00:00	62	43.4	18.6	2016-02-26 12:00:00	50.6	46.2	4.4
2015-02-16 0:00:00	50.2	29.5	20.7	2016-02-26 18:00:00	50.6	52.1	1.5
2015-02-16 6:00:00	50.2	45.3	4.9	2016-02-28 0:00:00	50.6	11.4	39.2
2015-02-16 12:00:00	50.2	38.0	12.2	2016-02-28 6:00:00	50.4	26.0	24.4
2015-02-16 18:00:00	50.2	32.0	18.2	2016-02-28 12:00:00	50.6	37.4	13.2
2015-03-09 18:00:00	54.2	49.5	4.7	2016-02-28 18:00:00	50.6	45.7	4.9

model presented a satisfactory performance for precipitations below 50 mm.

In order to evaluate the performance of temporal forecasting for the study region, Fig. 8 shows typical rainfall events in southeastern Brazil, during the Brazilian summer for the Oct/2014 - Oct/2016 period. The rainy season in the considered region typically begins at the end of October and lasts from 155 to 173 days (Minuzzi et al., 2007) ending approximately at the beginning of March. We observed that the highest concentration of critical extreme events (above 50 mm of precipitation) occurs between the months of December to February.

For the period considered, Table 9 shows the dates and times during which extreme events occurred. The record groups indicate 24-hour accumulated rainfall recorded every 6 h. Observing the in bold groups, it is possible to confirm that the best model (I4O1), with previous data of 24 h, predicts with greater precision the horizon of 6 h ahead. For example, for the group with date: 2014-12-13, time: 6:00:00 to date: 2014-12-14, time: 6:00:00, the prediction error continuously decreases reaching the value 0.8 mm. This same behavior occurs in 7 (54%) cases of the groups shown in Table 9. Based on these results, we can emphasize that the approach proposed in this work considers relevant atmospheric characteristics to predict the volume of precipitation in extreme event scenarios.

Finally, we compare our model with the Autoregressive Integrated Moving Average model - ARIMA. The model was built considering the same training data used in our approach (11,656 registers from Sep/2006 to October/2014). The best-fitted model was ARIMA(3,1,1) and the calculated coefficients are: $[ar1, ar2, ar3, ma1] = [0.4095, -0.0196, 0.0552, -0.9662]$. The ARIMA model achieved an RMSE value of 43.37 mm and MAE value of 21.62. Our best model I4O1 achieved during training an RMSE value of 0.94 mm. and MAE value of 0.46. It is important to highlight that during the operational test our model achieved an RMSE value of 6.94 mm and an MAE value of 2.97 mm, superior performance to that ARIMA model. This shows that our approach gives superior results to classical models based on time series.

Fig. 9 shows the responses of the I4O1 and ARIMA models for a time

window from Nov-28-2006, 00:00 to Dec-01-2006, 18:00, from training data. Considering that the precipitation forecast is carried out for 6-hours ahead, it is observed that the auto-regressive Integrated moving average models always have a delayed response, which is not of interest for the prediction of extreme events that would affect a city. The best model proposed in this work (I4O1) has predictive characteristics that allow it to track precipitation events without delay. This behavior is due to the fact that our approach takes into account several atmospheric variables, for different pressure levels, such as potential vorticity, specific humidity, vertical velocity, U velocity, V velocity, relative humidity, among other surface variables, which we relate through neural models to forecast rainfall. It is important to emphasize that the forecast of the volume of precipitation in rain is extremely difficult, due to the complexity of the meteorological system that produces them.

5. Conclusions and future directions

In this work we proposed a model based on LSTM neural networks to predict the volume of extreme precipitation and detect extreme rainfall events. The model uses data from 8 atmospheric variables and 11 surface variables, for 12 atmospheric pressure levels. We evaluated several configurations, and the model that obtained the best predictive accuracy uses 24 h prior information, before the target event.

We believe that this work contributes towards a new approach to predict extremes rainfall events in southeastern Brazil, especially in Belo Horizonte, and can lead to better predictability of extreme weather events. The approach presented here can be applied in other regions and locations, using the same data source, ERA-Interim. It would also be possible to use ERA5 (ERA5, 2020) data for training a similar forecast model. ERA5 covers the Earth in a 30 km grid and resolves the atmosphere using 137 surface levels, up to a height of 80 km. Our methodology is inherently objective and reproducible, and it can be implemented operationally to assist in the prediction of intense and potentially catastrophic events. This study shows that ERA-Interim as input data resulted in models with a good performance for estimating

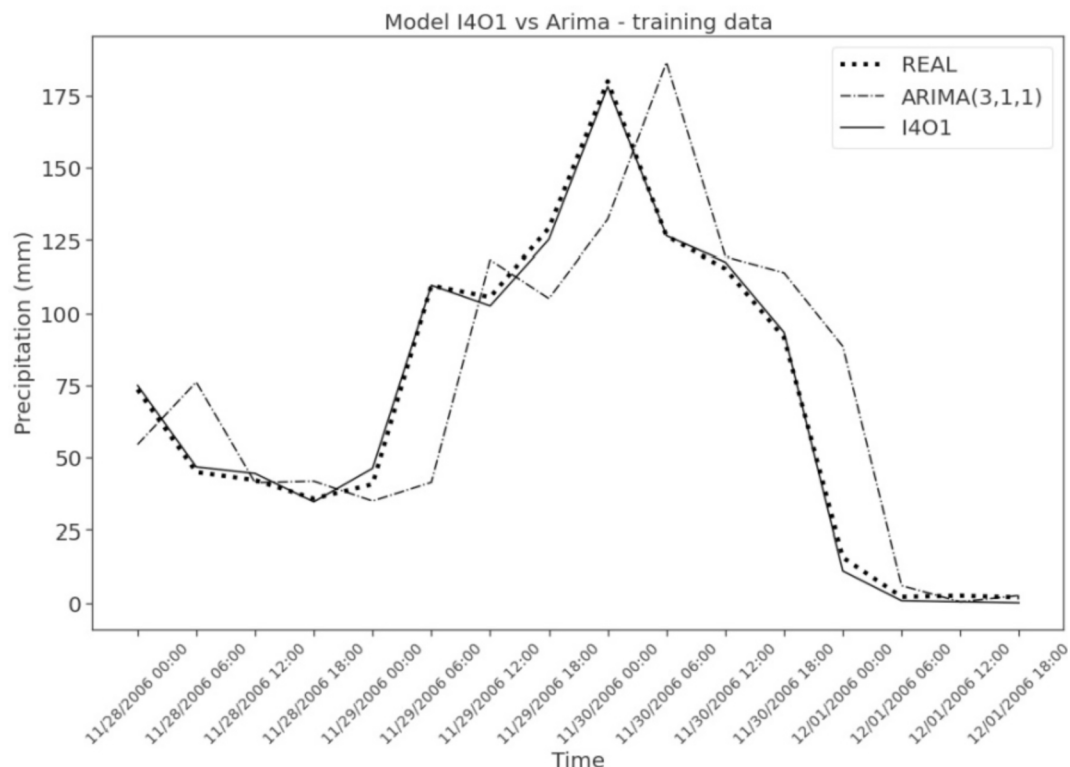


Fig. 9. Comparison of the response of the I4O1 and ARIMA models.

most events of heavy precipitation. Naturally, our data source can be replaced by other equivalent meteorological input data with real-time or near real-time measurements, and serve as support to the monitoring of heavy precipitations.

The vast majority of approaches treat extreme rain events forecasting as an imbalanced problem, such as the studies cited in our related works Section. Thus, the model requires a number of good positive-class examples that represent the extreme events and their anomalous conditions. The main contribution of our approach is that it is not necessary to separate scenarios of normal precipitation or absence of precipitation, which is a drawback of similar approaches. Our model is capable of handling a complete time series without additional preprocessing required.

The proposed methodology and the results of this work can be used to predict extreme precipitation events in the city of Belo Horizonte 24 h in advance. We highlight the following contributions: a) Possibility to be used as a decision support tool associated with the forecasts made using global mathematical models; b) The processing required is less computationally expensive than the usual meteorological forecasting strategies; c) The model's specialization by season can contribute to direct attention to specific neighborhoods and regions; and d) Flexibility to use various time windows for prediction, between 6 and 48 h.

As limitations of this study we highlight: a) Absence of spatial abstraction: the model needs to be retrained for use in a new area, and it is necessary to obtain and synchronize the data from the weather stations present in each target area, b) Dependent on a significant historical interval: the model used 8 years of historical data, and using smaller intervals might reduce its generalization capacity, and; c) Absence of previous studies describing strategies for predicting extreme rainfall events without balancing the data.

As a future work, we plan to experiment with other architectures and some methods of transforming non-stationary time-series data into stationary. We also suggest: a) considering a comparative performance analysis with different activation functions for LSTM networks, as recommend in Farzad et al. (2019). The authors establish that other activation functions such as Elliott (similar to sigmoid) can reduce average error and be faster to compute than the exponential sigmoid function; b) investigating new architectures, such as a convolutional neural networks (CNN), which can be used as the encoder in an encoder-decoder architecture, with the intention of capturing spatial abstraction (geolocation); c) replacing the ERA-Interim database with ERA-5 and observe how the models' results are affected. Note that ERA 5 can be used in order to reproduce our results, according to our methodology; d) using smaller temporal resolutions (the ERA-5 data has hourly resolution); e) applying the proposed methodology in other regions to investigate how much the methodology can be generalized for different environments, and; f) change the methodology to retrain the model whenever new daily data becomes available, effectively making it an online system.

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CRediT authorship contribution statement

André de Sousa Araújo: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Visualization. **Adma Raia Silva:** Conceptualization, Methodology. **Luis E. Zárate:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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